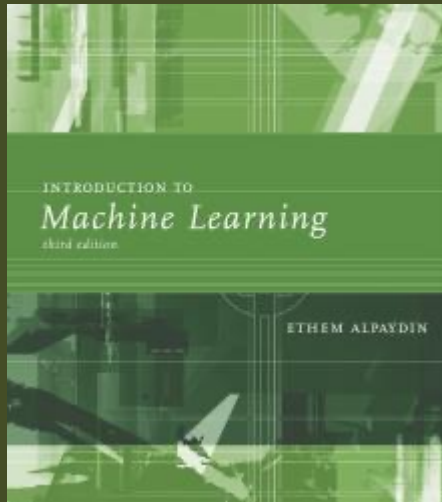




Lecture Slides for

INTRODUCTION TO MACHINE LEARNING

3RD EDITION

ETHEM ALPAYDIN

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CHAPTER 5:

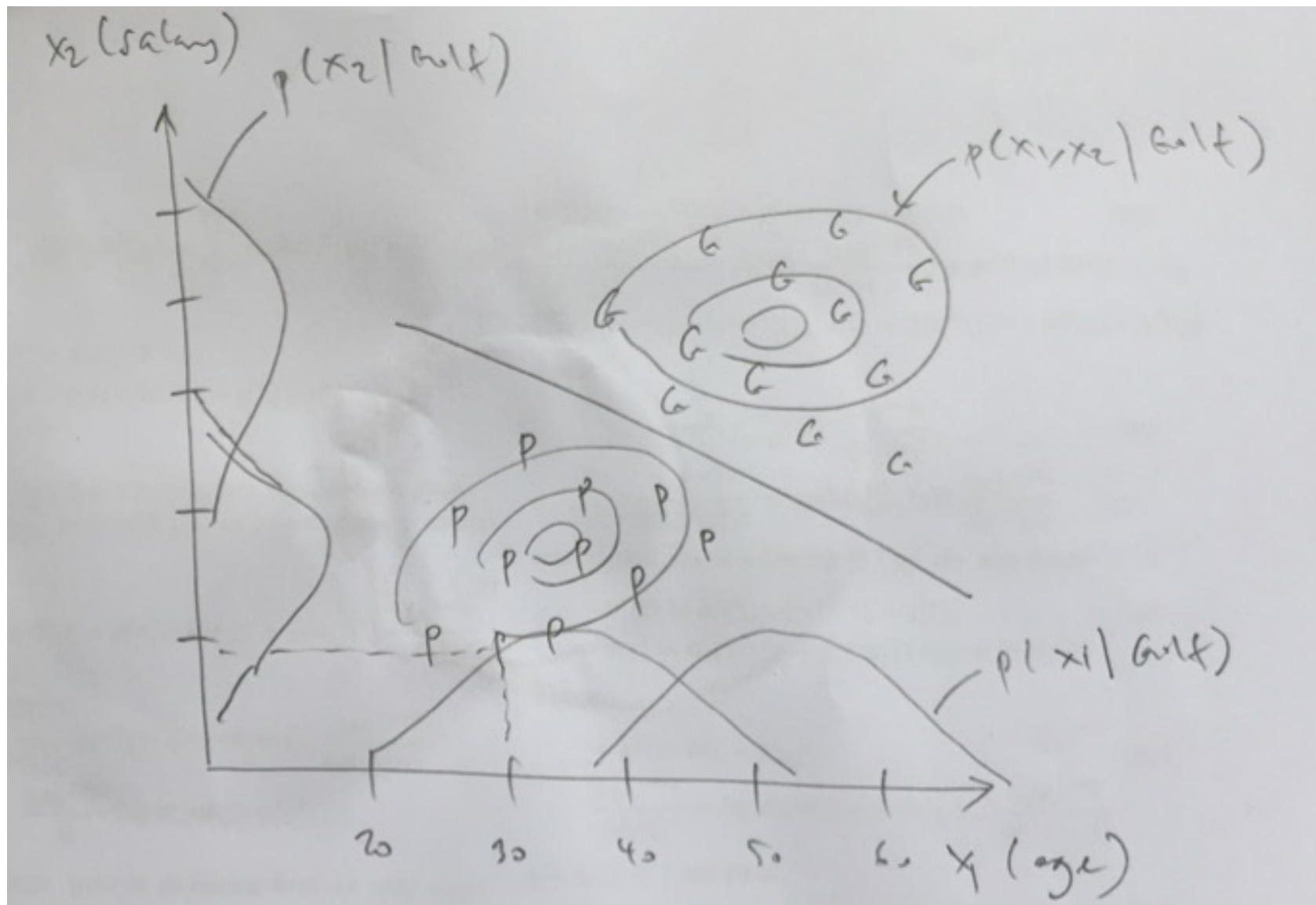
MULTIVARIATE METHODS

Multivariate Data

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- Multiple measurements (sensors)
- d inputs/features/attributes: d -variate
- N instances/observations/examples

$$\mathbf{X} = \begin{bmatrix} X_1^1 & X_2^1 & \dots & X_d^1 \\ X_1^2 & X_2^2 & \dots & X_d^2 \\ \vdots & & & \\ X_1^N & X_2^N & \dots & X_d^N \end{bmatrix}$$



Multivariate Parameters

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$$\text{Mean: } E[\mathbf{x}] = \boldsymbol{\mu} = [\mu_1, \dots, \mu_d]^T$$

$$\text{Covariance: } \sigma_{ij} \equiv \text{Cov}(X_i, X_j)$$

$$\text{Correlation: } \text{Corr}(X_i, X_j) \equiv \rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$$

$$\Sigma \equiv \text{Cov}(\mathbf{X}) = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T] = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2d} \\ \vdots & & & \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_d^2 \end{bmatrix}$$

Parameter Estimation

Sample mean $\mathbf{m}$: $m_i = \frac{\sum_{t=1}^N x_i^t}{N}, i = 1, \dots, d$

Covariance matrix $\mathbf{S}$: $s_{ij} = \frac{\sum_{t=1}^N (x_i^t - m_i)(x_j^t - m_j)}{N}$

Correlation matrix $\mathbf{R}$: $r_{ij} = \frac{s_{ij}}{s_i s_j}$

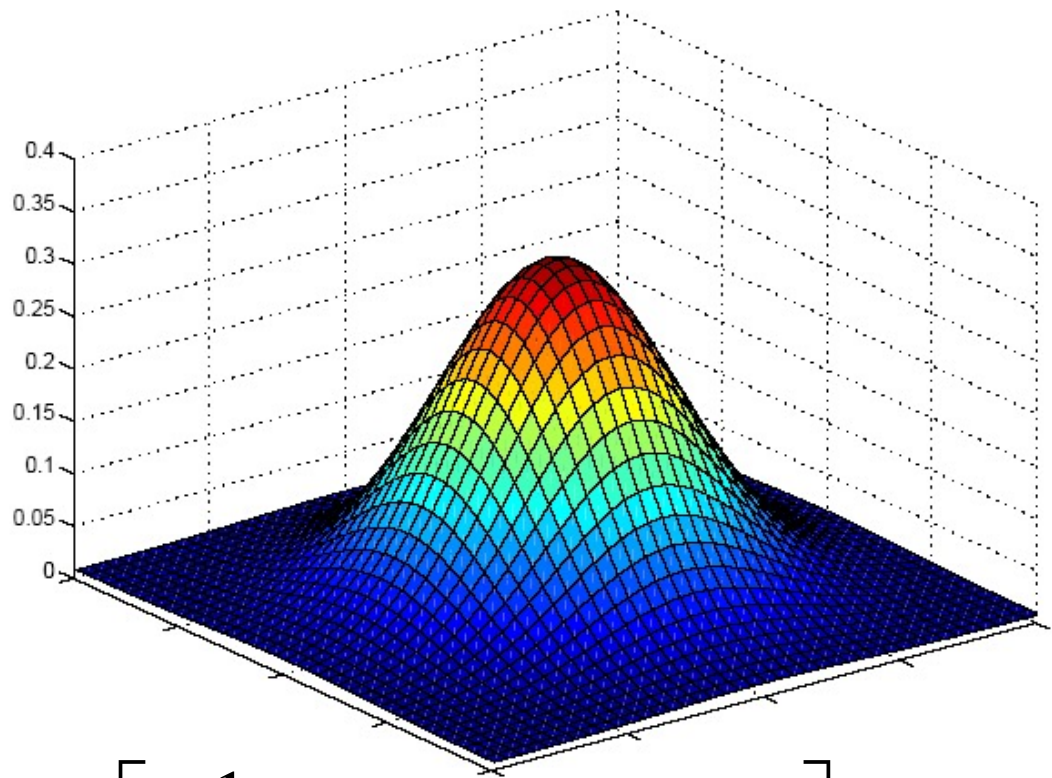
Estimation of Missing Values

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- What to do if certain instances have missing attributes?
- Ignore those instances: not a good idea if the sample is small
- Use 'missing' as an attribute: may give information
- **Imputation:** Fill in the missing value
 - ▣ Mean imputation: Use the most likely value (e.g., mean)
 - ▣ Imputation by regression: Predict based on other attributes

Multivariate Normal Distribution

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$$\mathbf{x} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

Multivariate Normal Distribution

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□ Mahalanobis distance: $(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$

measures the distance from $\mathbf{x}$ to $\boldsymbol{\mu}$ in terms of $\boldsymbol{\Sigma}$ (normalizes for difference in variances and correlations)

□ Bivariate: $d = 2$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

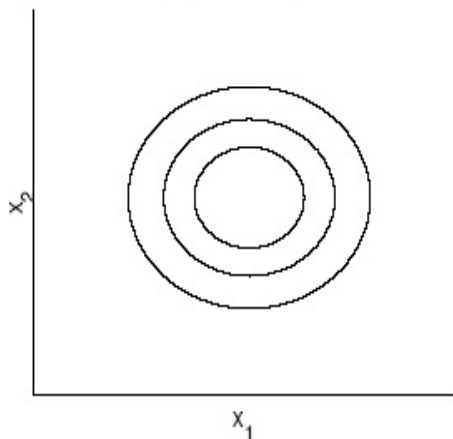
$$p(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}(z_1^2 - 2\rho z_1 z_2 + z_2^2)\right]$$

$$z_i = (x_i - \mu_i) / \sigma_i$$

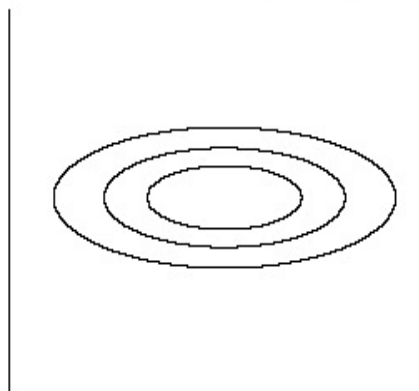
Bivariate Normal

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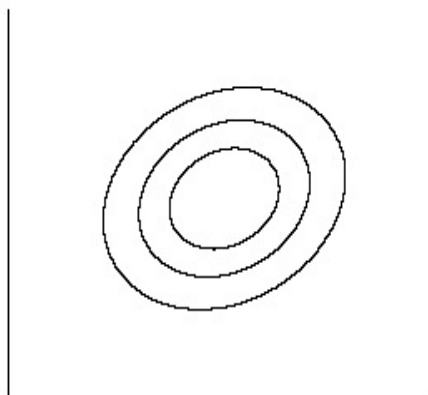
$$\text{Cov}(x_1, x_2) = 0, \text{Var}(x_1) = \text{Var}(x_2)$$



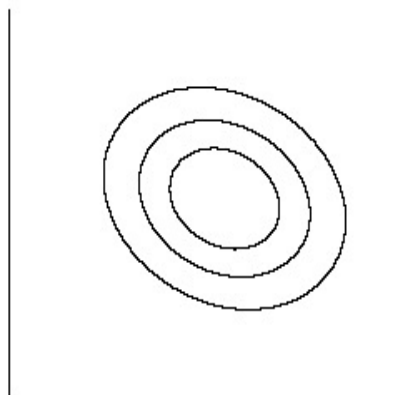
$$\text{Cov}(x_1, x_2) = 0, \text{Var}(x_1) > \text{Var}(x_2)$$



$$\text{Cov}(x_1, x_2) > 0$$

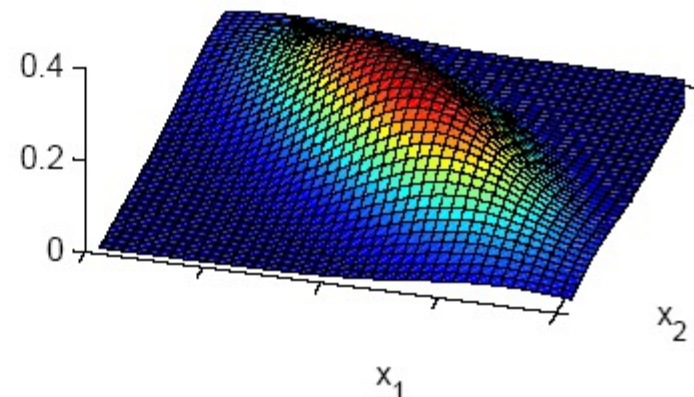
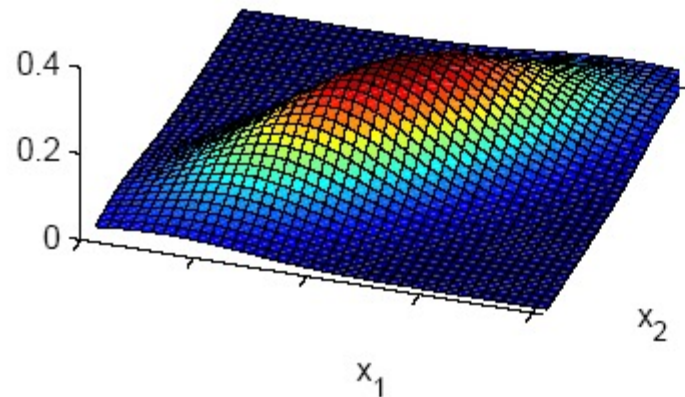
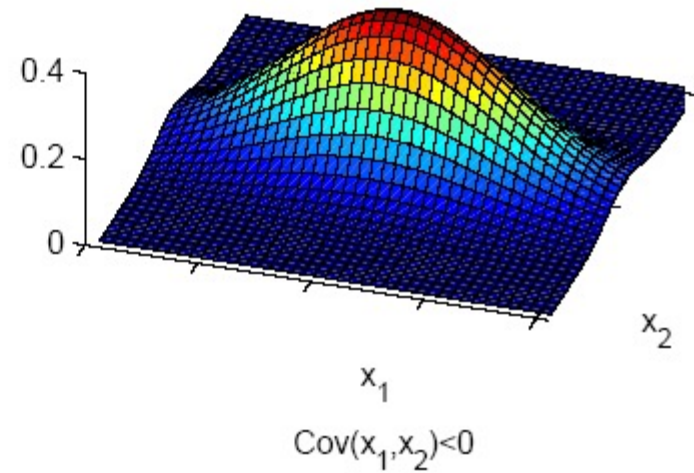
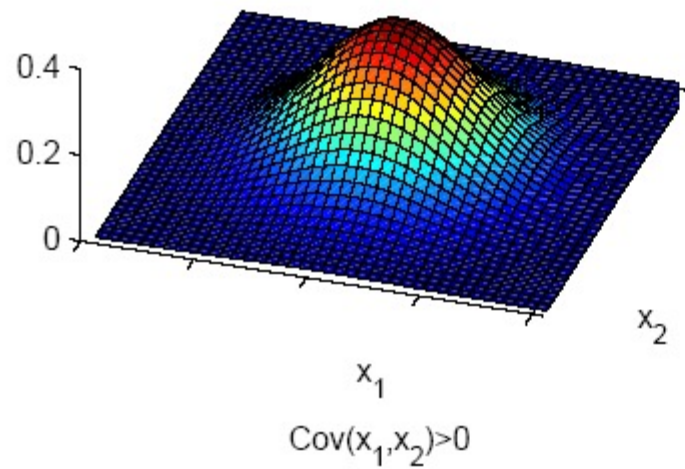


$$\text{Cov}(x_1, x_2) < 0$$



$$\text{Cov}(x_1, x_2) = 0, \text{Var}(x_1) = \text{Var}(x_2)$$

$$\text{Cov}(x_1, x_2) = 0, \text{Var}(x_1) > \text{Var}(x_2)$$



Independent Inputs: Naive Bayes

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- If x_i are independent, offdiagonals of Σ are 0, Mahalanobis distance reduces to weighted (by $1/\sigma_i$) Euclidean distance:

$$p(\mathbf{x}) = \prod_{i=1}^d p_i(x_i) = \frac{1}{(2\pi)^{d/2} \prod_{i=1}^d \sigma_i} \exp \left[-\frac{1}{2} \sum_{i=1}^d \left(\frac{x_i - \mu_i}{\sigma_i} \right)^2 \right]$$

- If variances are also equal, reduces to Euclidean distance

Parametric Classification

- If $p(\mathbf{x} | C_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$

$$p(\mathbf{x} | C_i) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}_i|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right]$$

- Discriminant functions

$$\begin{aligned} g_i(\mathbf{x}) &= \log p(\mathbf{x} | C_i) + \log P(C_i) \\ &= -\frac{d}{2} \log 2\pi - \frac{1}{2} \log |\boldsymbol{\Sigma}_i| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) + \log P(C_i) \end{aligned}$$

Estimation of Parameters

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$$\hat{P}(C_i) = \frac{\sum_t r_i^t}{N}$$

$$\mathbf{m}_i = \frac{\sum_t r_i^t \mathbf{x}^t}{\sum_t r_i^t}$$

$$\mathbf{S}_i = \frac{\sum_t r_i^t (\mathbf{x}^t - \mathbf{m}_i)(\mathbf{x}^t - \mathbf{m}_i)^T}{\sum_t r_i^t}$$

$$g_i(\mathbf{x}) = -\frac{1}{2} \log |\mathbf{S}_i| - \frac{1}{2} (\mathbf{x} - \mathbf{m}_i)^T \mathbf{S}_i^{-1} (\mathbf{x} - \mathbf{m}_i) + \log \hat{P}(C_i)$$

Different $\mathbf{S}_i$

□ Quadratic discriminant

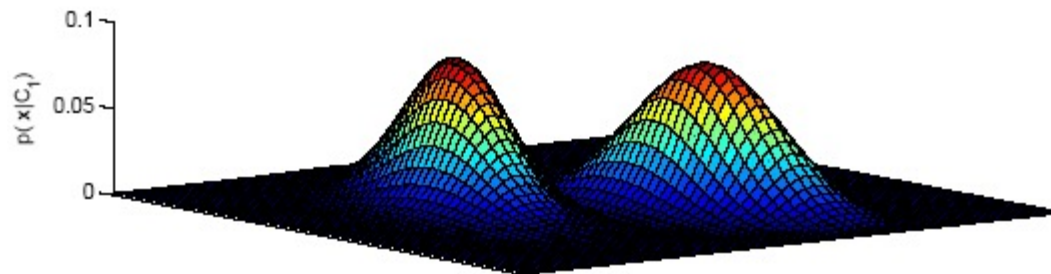
$$\begin{aligned} g_i(\mathbf{x}) &= -\frac{1}{2} \log |\mathbf{S}_i| - \frac{1}{2} (\mathbf{x}^T \mathbf{S}_i^{-1} \mathbf{x} - 2 \mathbf{x}^T \mathbf{S}_i^{-1} \mathbf{m}_i + \mathbf{m}_i^T \mathbf{S}_i^{-1} \mathbf{m}_i) + \log \hat{P}(C_i) \\ &= \mathbf{x}^T \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^T \mathbf{x} + w_{i0} \end{aligned}$$

where

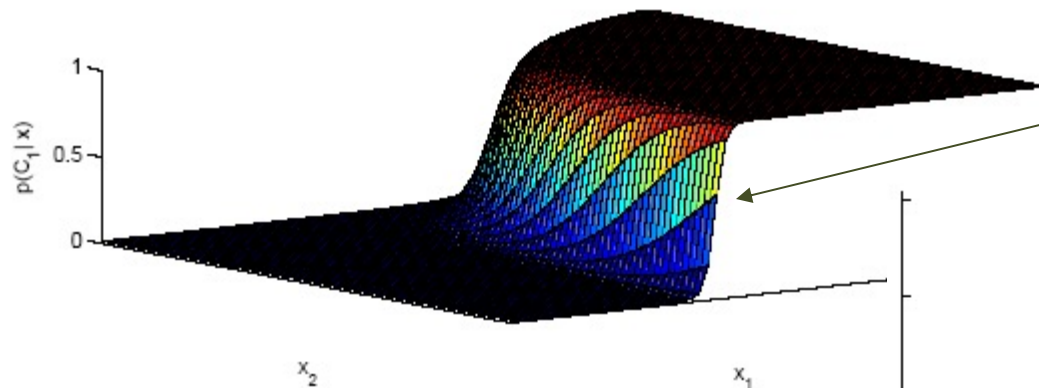
$$\mathbf{W}_i = -\frac{1}{2} \mathbf{S}_i^{-1}$$

$$\mathbf{w}_i = \mathbf{S}_i^{-1} \mathbf{m}_i$$

$$w_{i0} = -\frac{1}{2} \mathbf{m}_i^T \mathbf{S}_i^{-1} \mathbf{m}_i - \frac{1}{2} \log |\mathbf{S}_i| + \log \hat{P}(C_i)$$

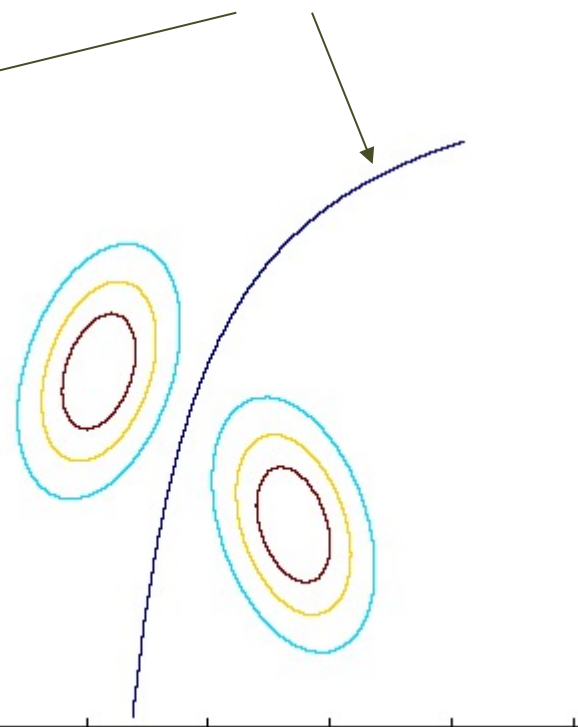


likelihoods



posterior for C_1

discriminant:
 $P(C_1 | \mathbf{x}) = 0.5$



Common Covariance Matrix $\mathbf{S}$

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- Shared common sample covariance $\mathbf{S}$

$$\mathbf{S} = \sum_i \hat{P}(C_i) \mathbf{S}_i$$

- Discriminant reduces to

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \mathbf{m}_i)^T \mathbf{S}^{-1}(\mathbf{x} - \mathbf{m}_i) + \log \hat{P}(C_i)$$

which is a linear discriminant

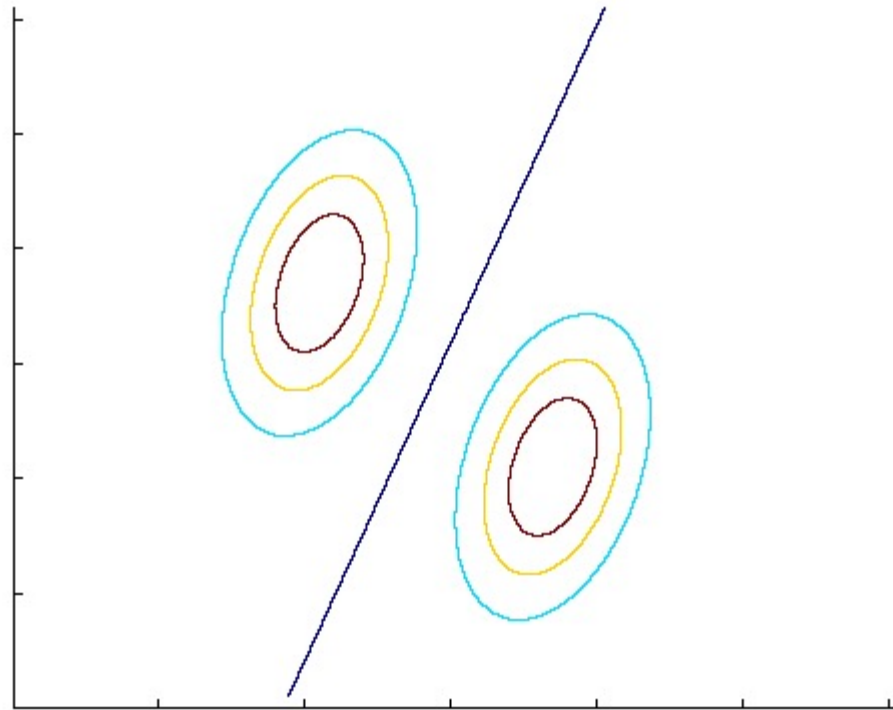
$$g_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} + w_{i0}$$

where

$$\mathbf{w}_i = \mathbf{S}^{-1} \mathbf{m}_i \quad w_{i0} = -\frac{1}{2} \mathbf{m}_i^T \mathbf{S}^{-1} \mathbf{m}_i + \log \hat{P}(C_i)$$

Common Covariance Matrix $\mathbf{S}$

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Diagonal Σ

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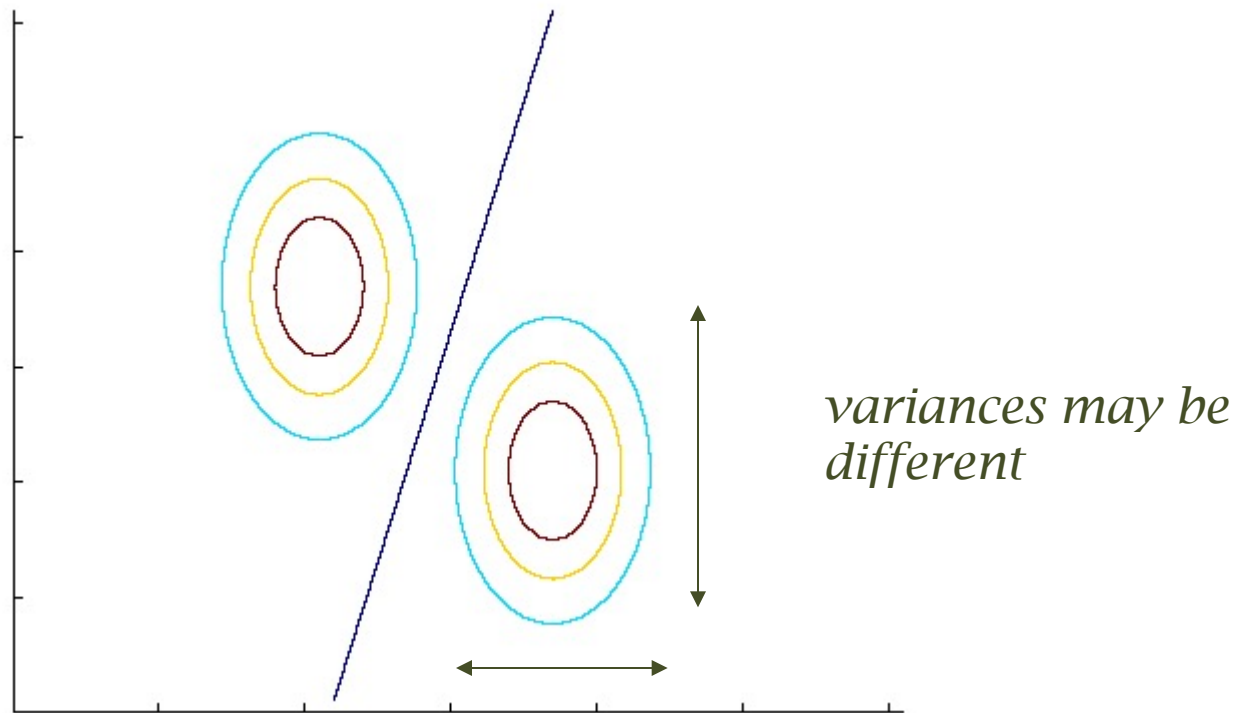
- When $x_j, j = 1, \dots, d$, are independent, Σ is diagonal
 $p(\mathbf{x} | C_i) = \prod_j p(x_j | C_i)$ (Naive Bayes' assumption)

$$g_i(\mathbf{x}) = -\frac{1}{2} \sum_{j=1}^d \left(\frac{x_j^t - m_{ij}}{s_j} \right)^2 + \log \hat{P}(C_i)$$

Classify based on weighted Euclidean distance (in s_j units) to the nearest mean

Diagonal $\mathbf{S}$

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Diagonal $\mathbf{S}$, equal variances

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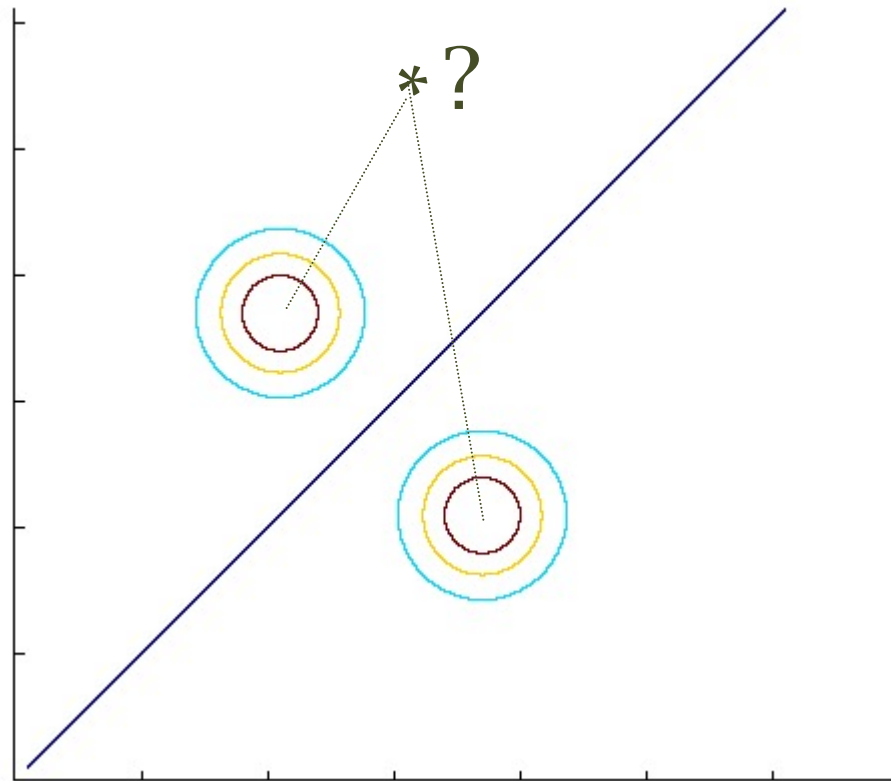
- Nearest mean classifier: Classify based on Euclidean distance to the nearest mean

$$\begin{aligned} g_i(\mathbf{x}) &= -\frac{\|\mathbf{x} - \mathbf{m}_i\|^2}{2s^2} + \log \hat{P}(C_i) \\ &= -\frac{1}{2s^2} \sum_{j=1}^d (x_j^t - m_{ij})^2 + \log \hat{P}(C_i) \end{aligned}$$

- Each mean can be considered a prototype or template and this is template matching

Diagonal $\mathbf{S}$, equal variances

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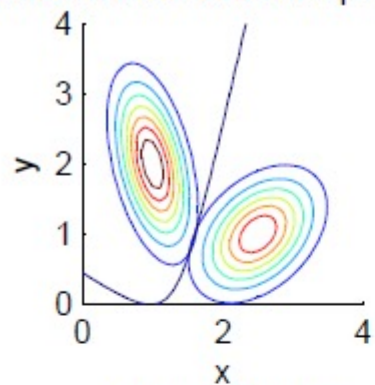
Model Selection

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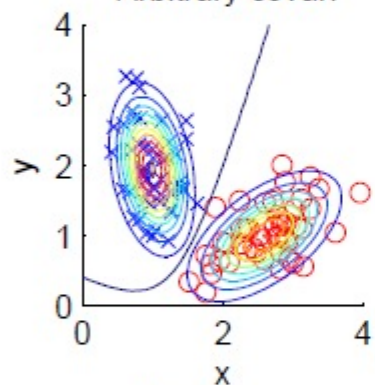
<i>Assumption</i>	<i>Covariance matrix</i>	<i>No of parameters</i>
Shared, Hyperspheric	$\mathbf{S}_i = \mathbf{S} = s^2 \mathbf{I}$	1
Shared, Axis-aligned	$\mathbf{S}_i = \mathbf{S}$, with $s_{ij} = 0$	d
Shared, Hyperellipsoidal	$\mathbf{S}_i = \mathbf{S}$	$d(d+1)/2$
Different, Hyperellipsoidal	$\mathbf{S}_i$	$K d(d+1)/2$

- As we increase complexity (less restricted $\mathbf{S}$), bias decreases and variance increases
- Assume simple models (allow some bias) to control variance (regularization)

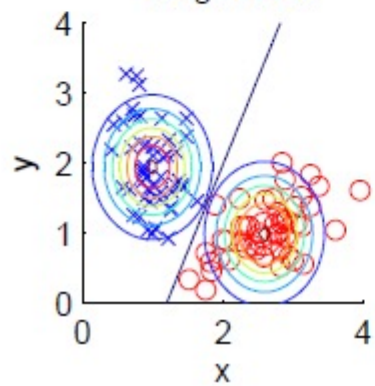
Population likelihoods and posteriors



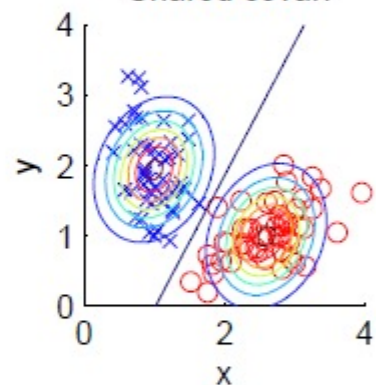
Arbitrary covar.



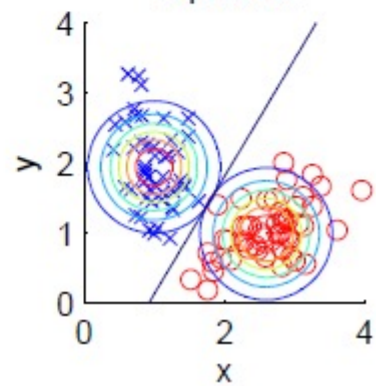
Diag. covar.



Shared covar.



Equal var.



Discrete Features

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- Binary features: $p_{ij} \equiv p(x_j=1|C_i)$
if x_j are independent (Naive Bayes')

$$p(\mathbf{x} | C_i) = \prod_{j=1}^d p_{ij}^{x_j} (1 - p_{ij})^{(1-x_j)}$$

the discriminant is linear

$$\begin{aligned} g_i(\mathbf{x}) &= \log p(\mathbf{x} | C_i) + \log P(C_i) \\ &= \sum_j [x_j \log p_{ij} + (1 - x_j) \log (1 - p_{ij})] + \log P(C_i) \end{aligned}$$

Estimated parameters

$$\hat{p}_{ij} = \frac{\sum_t x_j^t r_i^t}{\sum_t r_i^t}$$

Discrete Features

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- Multinomial (1-of- n_j) features: $x_j \in \{v_1, v_2, \dots, v_{n_j}\}$

$$p_{ijk} \equiv p(z_{jk}=1 | C_i) = p(x_j = v_k | C_i)$$

if x_j are independent

$$p(\mathbf{x} | C_i) = \prod_{j=1}^d \prod_{k=1}^{n_j} p_{ijk}^{z_{jk}}$$

$$g_i(\mathbf{x}) = \sum_j \sum_k z_{jk} \log p_{ijk} + \log P(C_i)$$

$$\hat{p}_{ijk} = \frac{\sum_t z_{jk}^t r_i^t}{\sum_t r_i^t}$$

Multivariate Regression

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$$r^t = g(x^t | w_0, w_1, \dots, w_d) + \varepsilon$$

Multivariate linear model

$$w_0 + w_1 x_1^t + w_2 x_2^t + \dots + w_d x_d^t$$

$$E(w_0, w_1, \dots, w_d | \mathcal{X}) = \frac{1}{2} \sum_t [r^t - w_0 - w_1 x_1^t - \dots - w_d x_d^t]^2$$

Multivariate polynomial model:

Define new higher-order variables

$$z_1 = x_1, z_2 = x_2, z_3 = x_1^2, z_4 = x_2^2, z_5 = x_1 x_2$$

and use the linear model in this new $\mathbf{z}$ space

(basis functions, kernel trick: Chapter 13)